

Lecture 1

Matrix Lie groups and their Lie algebras

Objectives. *Part II begins. We define matrix Lie groups — closed subgroups of $GL(n, \mathbb{C})$ — and develop the matrix exponential, using it to attach to every matrix Lie group its Lie algebra; we compute these algebras for the classical groups, prove they are closed under the commutator, and establish in what sense the exponential bridges the two structures: a perfect dictionary locally, a subtler one globally. Complexification and the physicist's Hermitian generators close the lecture.*

1.1 Matrix space, and the definition of a matrix Lie group

Part I studied symmetry in finite supply: groups whose elements can be listed, summed over, averaged. The groups of Part II — rotations through every angle, boosts to every velocity, phases of every argument — form continua, and the single new capability this brings is the one this lecture is built around: *differentiation*. Differentiating a continuous group at its identity produces a linear object, the *Lie algebra*, on which nearly every computation of Part II will be performed; the *matrix exponential* climbs back up. This lecture constructs both ends of the bridge and crosses it in both directions.

Notation. The operator a representation assigns to a group element, written $\rho(g)$ in Part I, is written $\Pi(g)$ from here on (where it usually carries a continuous label); the two symbols denote the same object, and the function action $(\Pi(g)\psi)(x) = \psi(g^{-1}x)$ of Lecture 3 is unchanged.

Everything happens inside the space $M_n(\mathbb{C})$ of complex $n \times n$ matrices, the vector space $\mathbb{C}^{n^2}$ in a matrix costume, measured with the *operator norm* $\|X\| = \sup_{v \neq 0} |Xv|/|v|$. The norm obeys the triangle inequality and, crucially, is *submultiplicative*, $\|XY\| \leq \|X\| \|Y\|$; every eigenvalue of X obeys $|\lambda| \leq \|X\|$. All norms on a finite-dimensional space are equivalent (analysis), so convergence in $\|\cdot\|$ is entrywise convergence, $M_n(\mathbb{C})$ is complete, and compact means closed and bounded (Heine–Borel). The algebra respects the topology: multiplication and det are polynomial in the entries, hence continuous, so

$$GL(n, \mathbb{C}) = \det^{-1}(\mathbb{C} \setminus \{0\})$$

is an *open* subset of $M_n(\mathbb{C})$, on which inversion is continuous (Cramer's formula). No topology beyond such metric-space language will be needed.

Definition 1.1 (Matrix Lie group). A *matrix Lie group* is a subgroup G of $GL(n, \mathbb{C})$ that is closed as a subset of $GL(n, \mathbb{C})$: whenever $A_m \in G$ converge to a matrix A with $\det A \neq 0$, the limit A lies in G .

Closedness is relative to $\mathrm{GL}(n, \mathbb{C})$, not $M_n(\mathbb{C})$: sequences in G may converge to singular matrices, about which the definition is silent. Thus $\mathrm{GL}(n, \mathbb{C})$ itself qualifies, although $\mathrm{diag}(1/m, 1, \dots, 1)$ escapes it in the limit; and since entrywise reality is a closed condition, $\mathrm{GL}(n, \mathbb{R})$ qualifies too. Why insist on closedness? Because subgroups of $\mathrm{GL}(n, \mathbb{C})$ can be strange in a way no symmetry group of physics is.

Example 1.2 (A line wound densely on a torus). Fix α irrational and let

$$G_\alpha = \left\{ \begin{pmatrix} e^{it} & 0 \\ 0 & e^{i\alpha t} \end{pmatrix} : t \in \mathbb{R} \right\} \subset \mathrm{GL}(2, \mathbb{C}),$$

a respectable subgroup: the line $\mathbb{R}$ wound with slope α around the torus $\mathrm{U}(1) \times \mathrm{U}(1)$ of diagonal unitaries. The winding never closes up, and the closure of G_α is the *whole* torus (Sheet 8): a one-parameter family whose limits fill a two-parameter family. Closedness excludes exactly this pathology — and it legitimises the limit arguments from which every theorem of this lecture springs.

Remark 1.3 (Prerequisites, quoted freely). Part II draws on the prerequisites at full strength: from analysis, the Heine–Borel theorem and (once, in Theorem 1.31) the inverse function theorem; from linear algebra, the Schur triangularisation and the spectral normal forms of unitary and of special orthogonal matrices. Each use is flagged; everything else is proved.

1.2 The zoo: closedness, compactness, connectedness

The matrix groups of Lecture 1 return as protagonists, and one pattern verifies them all: each is carved out of $\mathrm{GL}(n, \mathbb{C})$ by equations $\Phi(A) = \text{const}$ with Φ continuous, and solution sets of continuous equations are closed.

Proposition 1.4 (The classical groups are matrix Lie groups). $\mathrm{GL}(n, \mathbb{R})$, $\mathrm{SL}(n, \mathbb{R})$, $\mathrm{SL}(n, \mathbb{C})$, $\mathrm{O}(n)$, $\mathrm{SO}(n)$, $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ are matrix Lie groups.

Proof. Each is a subgroup (Lecture 1), cut out of $\mathrm{GL}(n, \mathbb{C})$ by conditions among $\mathrm{Im} A = 0$, $A^\top A = \mathbb{I}$, $A^\dagger A = \mathbb{I}$, $\det A = 1$: finitely many closed conditions. ■

Two new members join the zoo; both are previews.

Example 1.5 (The Lorentz group). With $\eta = \mathrm{diag}(1, -1, -1, -1)$, the Lorentz group

$$\mathrm{O}(1, 3) = \{\Lambda \in \mathrm{GL}(4, \mathbb{R}) : \Lambda^\top \eta \Lambda = \eta\}$$

consists of the linear maps preserving the Minkowski form $t^2 - x^2 - y^2 - z^2$; it is a matrix Lie group by the same pattern. Unlike its Euclidean cousin $\mathrm{O}(4)$ it is unbounded, containing for every $\chi \in \mathbb{R}$ the *boost* $\Lambda(\chi) = \begin{pmatrix} \cosh \chi & \sinh \chi \\ \sinh \chi & \cosh \chi \end{pmatrix} \oplus \mathbb{I}_2$. Lecture 14 is devoted to this group.

Example 1.6 (The Heisenberg group). The Heisenberg group H_3 of unitriangular matrices $\begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$, $a, b, c \in \mathbb{R}$, is closed (fixed entries are closed conditions) and unbounded. Multiplication adds the parameters a, b but twists the corner, $c \mapsto c + c' + ab'$: here, Lecture 13 will show, hide the canonical commutation relations of quantum mechanics. Note the device — *translations* made matrix-shaped by padding a dimension — reused in Lectures 13 and 15. (The symplectic family enters on Sheet 8 via $\mathrm{Sp}(2, \mathbb{R}) = \mathrm{SL}(2, \mathbb{R})$.)

Example 1.7 (SU(2) is a sphere). For $U \in \text{SU}(2)$, $\det U = 1$ makes the inverse the adjugate; equating it with $U^\dagger$ entry by entry leaves

$$\text{SU}(2) = \left\{ \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} : \alpha, \beta \in \mathbb{C}, |\alpha|^2 + |\beta|^2 = 1 \right\}, \quad (1.1)$$

the unit sphere S^3 in $\mathbb{C}^2 \cong \mathbb{R}^4$: compact at a glance, connected, and — being a sphere — without holes that a loop could detect, a property Lecture 9 will name *simply connected* and exploit heavily.

Proposition 1.8 (Compactness in the zoo). $\text{O}(n)$, $\text{SO}(n)$, $\text{U}(n)$, $\text{SU}(n)$ are compact; GL and SL over $\mathbb{R}$ or $\mathbb{C}$ (for $n \geq 2$), $\text{O}(1, 3)$ and H_3 are not.

Proof. A unitary preserves lengths, so $\|U\| = 1$: $\text{U}(n)$ is bounded; and it is closed in all of $M_n(\mathbb{C})$, since $A^\dagger A = \mathbb{I}$ is a closed condition that itself forces invertibility. Closed and bounded means compact, and $\text{O}(n)$, $\text{SO}(n)$, $\text{SU}(n)$ are closed subsets of $\text{U}(n)$. For the rest, unbounded families: $\text{diag}(t, 1/t, 1, \dots, 1)$, the boosts $\Lambda(\chi)$, and $a \rightarrow \infty$ in H_3 . ■

Compactness will become a load-bearing wall in Lecture 12: it is what lets Part I's averaging trick survive, with sums replaced by integrals. The other geographic feature is connectedness: a set is *path-connected* if any two of its points are joined by a continuous path inside it, *connected* if it admits no partition into two nonempty relatively open pieces; paths imply connectedness (analysis), and the converse for matrix Lie groups falls out of Section 1.5.

Proposition 1.9 ($\text{O}(n)$ is disconnected). $\text{O}(n)$ is not connected; nor is $\text{GL}(n, \mathbb{R})$.

Proof. On $\text{O}(n)$, $(\det A)^2 = 1$ and the continuous $\det$ attains both values ± 1 (at $\mathbb{I}$ and at $\text{diag}(-1, 1, \dots, 1)$): the sets $\{\det = \pm 1\}$ are nonempty, open in $\text{O}(n)$, and partition it. For $\text{GL}(n, \mathbb{R})$, use the sign of $\det$. ■

So $\text{O}(n)$ splits into $\text{SO}(n)$ and its reflection coset — the continuous shadow of the rotation/reflection split in Lecture 1's dihedral groups. The positive assertions will be earned: $\text{U}(n)$, $\text{SU}(n)$, $\text{SO}(n)$ in Section 1.5; $\text{GL}(n, \mathbb{C})$, $\text{SL}(n, \mathbb{C})$, $\text{SL}(n, \mathbb{R})$ on Sheet 8; the Lorentz group has exactly *four* pieces (Lecture 14) — why parity and time reversal are discrete choices, not limits of rotations.

Remark 1.10 (Abstract Lie groups). The wider literature defines a Lie group as a group that is also a smooth manifold — geometry we have not assumed. Nothing is lost: by a theorem of É. Cartan every closed subgroup of $\text{GL}(n, \mathbb{C})$ is such an object, and almost every Lie group of physical relevance is a matrix group or a quotient of one (the universal cover of $\text{SL}(2, \mathbb{R})$ is the standard exception). Here, “Lie group” means matrix Lie group.

1.3 The matrix exponential

The tool that turns linear data into group elements is the oldest function in analysis, promoted to matrices.

Definition 1.11 (Matrix exponential). For $X \in M_n(\mathbb{C})$,

$$e^X = \exp X = \sum_{k=0}^{\infty} \frac{X^k}{k!}. \quad (1.2)$$

Submultiplicativity dominates the series by $\sum \|X\|^k / k! = e^{\|X\|}$, so (1.2) converges absolutely ($M_n(\mathbb{C})$ is complete), uniformly on bounded sets (Weierstrass), and $\exp$ is continuous with $\|e^X\| \leq e^{\|X\|}$.

Proposition 1.12 (Calculus of the exponential). Let $X, Y \in M_n(\mathbb{C})$, $g \in \mathrm{GL}(n, \mathbb{C})$, $s, t \in \mathbb{C}$.

- (i) $e^0 = \mathbb{I}$; $(e^X)^\top = e^{X^\top}$, $(e^X)^\dagger = e^{X^\dagger}$.
- (ii) $e^{gXg^{-1}} = g e^X g^{-1}$; consequently $X = P \operatorname{diag}(\lambda_i) P^{-1}$ gives $e^X = P \operatorname{diag}(e^{\lambda_i}) P^{-1}$.
- (iii) If $XY = YX$ then $e^{X+Y} = e^X e^Y$.
- (iv) $e^{(s+t)X} = e^{sX} e^{tX}$; in particular e^X is invertible with inverse e^{-X} , so $\exp$ maps $M_n(\mathbb{C})$ into $\mathrm{GL}(n, \mathbb{C})$.
- (v) $\frac{d}{dt} e^{tX} = X e^{tX} = e^{tX} X$.

Proof. (i) holds term by term. (ii): $(gXg^{-1})^k = gX^k g^{-1}$ and conjugation is continuous. (iii): absolute convergence permits the Cauchy product,

$$e^X e^Y = \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^n \binom{n}{k} X^k Y^{n-k} = \sum_{n=0}^{\infty} \frac{(X+Y)^n}{n!},$$

the binomial theorem in the last step being valid precisely because X, Y commute — and *only* then: the nilpotent pair E_{12}, E_{21} already has $e^X e^Y \neq e^{X+Y}$, visible in the traces alone, since $e^{E_{12}} e^{E_{21}} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ has trace 3 whereas $e^{E_{12}+E_{21}} = e^{\sigma_1}$ has trace $2 \cosh 1 \approx 3.086$. (iv) is (iii) for sX, tX ; take $t = -s = 1$ for the inverse. (v): the term-by-term derivative $\sum t^k X^{k+1} / k!$ converges locally uniformly in t , justifying the differentiation and factoring as stated. ■

Example 1.13 (The circle, twice). Let $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, so $J^2 = -\mathbb{I}$ and the even/odd split of the series gives

$$e^{\theta J} = \cos \theta \mathbb{I} + \sin \theta J = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = R(\theta) :$$

all of $\mathrm{SO}(2)$ flows out of the single matrix J . Identifying $\mathbb{R}^2$ with $\mathbb{C}$ turns the computation into $e^{i\theta} = \cos \theta + i \sin \theta$: Euler's formula is a statement about $\mathrm{U}(1) \cong \mathrm{SO}(2)$ (Lecture 2), with J as multiplication by i . Figure 1.1 shows the picture that organises this whole lecture.

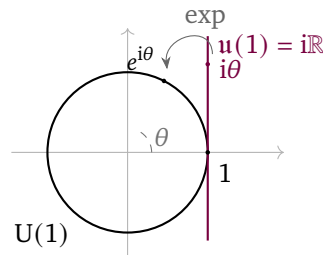


Figure 1.1: The exponential map of the circle group: $\exp$ wraps the tangent line $u(1) = i\mathbb{R}$ around $\mathrm{U}(1)$ with period 2π — bijective near the identity, not injective globally, surjective onto the connected compact group.

Example 1.14 (The exponential of $\mathfrak{su}(2)$, in closed form). Introduce the *Pauli matrices*

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

Hermitian, traceless, and satisfying (direct multiplication)

$$\sigma_j \sigma_k = \delta_{jk} \mathbb{K} + i \epsilon_{jkl} \sigma_l. \quad (1.3)$$

For a unit vector $\vec{n}$, contracting (1.3) with $n_j n_k$ kills the antisymmetric part, so $(\vec{n} \cdot \vec{\sigma})^2 = \mathbb{K}$ and the even/odd split gives

$$e^{-i\frac{\theta}{2} \vec{n} \cdot \vec{\sigma}} = \cos \frac{\theta}{2} \mathbb{K} - i \sin \frac{\theta}{2} \vec{n} \cdot \vec{\sigma}, \quad (1.4)$$

which has the form (1.1): these exponentials lie in $SU(2)$, and Section 1.5 will show they exhaust it. No equation will be used more often in Part II. The halved angle is no affectation: $\theta = 2\pi$ yields $-\mathbb{K}$, the first fingerprint of the double cover unmasked in Lecture 11.

Theorem 1.15 (Spectral mapping; determinant and trace). *If X has eigenvalues $\lambda_1, \dots, \lambda_n$ (with multiplicity), then e^X has eigenvalues $e^{\lambda_1}, \dots, e^{\lambda_n}$, and*

$$\det e^X = e^{\text{tr} X}. \quad (1.5)$$

Proof. Schur (Remark 1.3): $X = VTV^\dagger$, V unitary, T upper triangular with diagonal $\lambda_1, \dots, \lambda_n$. Powers of triangular matrices are triangular with powered diagonals, so e^T is triangular with diagonal e^{λ_j} — its eigenvalues, and by (ii) of Proposition 1.12 those of e^X ; finally $\det e^X = \prod_j e^{\lambda_j} = e^{\text{tr} X}$. ■

Equation (1.5) is the first entry of the group–algebra dictionary: the multiplicative condition $\det A = 1$ corresponds to the linear condition $\text{tr} X = 0$. Next, the exponential is inverted.

Theorem 1.16 (exp and log are local inverses). *For $\|A - \mathbb{K}\| < 1$ define*

$$\log A = \sum_{k \geq 1} (-1)^{k+1} \frac{(A - \mathbb{K})^k}{k},$$

absolutely and locally uniformly convergent, hence continuous. Then:

- (i) $e^{\log A} = A$ whenever $\|A - \mathbb{K}\| < 1$;
- (ii) $\log e^X = X$ whenever $\|X\| < \log 2$ (and then $\|e^X - \mathbb{K}\| \leq e^{\|X\|} - 1 < 1$).

In particular exp is injective on $\{\|X\| < \log 2\}$.

Proof. Both are scalar identities promoted by density and continuity. (i) For diagonalisable $A = P \text{diag}(d_j) P^{-1}$: each $|d_j - 1| \leq \|A - \mathbb{K}\| < 1$ (eigenvalue bound), both series pass through $P(\cdot)P^{-1}$ termwise, and $e^{\log d_j} = d_j$ on $|z - 1| < 1$ (analysis). General A are limits of diagonalisable A_m — perturb the diagonal of a Schur form $VTV^\dagger$ to pairwise distinct entries, unitary conjugation preserving the norm — so pass to the limit by continuity of exp and log. (ii) The estimate is the termwise bound; the identity follows by the same two steps from $\log e^z = z$ for $|z| < \log 2$, which keeps $|e^z - 1| < 1$. Injectivity: apply log to $e^X = e^Y$. ■

Two innocent-looking theorems can now be harvested; they carry most of Part II on their shoulders.

Theorem 1.17 (One-parameter subgroups). *Every continuous homomorphism $\gamma: (\mathbb{R}, +) \rightarrow \text{GL}(n, \mathbb{C})$ is of the form $\gamma(t) = e^{tX}$ for a unique $X \in M_n(\mathbb{C})$; in particular γ is automatically differentiable, and $X = \gamma'(0)$.*

Proof. Set $\varepsilon = \frac{1}{2} \log 2$, so $\exp$ is injective on $B = \{\|Z\| < 2\varepsilon\}$. By continuity at 0 there is $t_0 > 0$ with $\gamma(t) \in \exp(\{\|Z\| < \varepsilon\})$ for $|t| \leq t_0$; put $X = \frac{1}{t_0} \log \gamma(t_0)$, so $\gamma(t_0) = e^{t_0 X}$ with $\|t_0 X\| < \varepsilon$. Write $\gamma(t_0/2) = e^Y$, $\|Y\| < \varepsilon$; then $e^{2Y} = \gamma(t_0/2)^2 = \gamma(t_0) = e^{t_0 X}$ with both exponents in B , where $\exp$ is injective: $Y = \frac{t_0}{2} X$. Iterating halvings and using the homomorphism property, $\gamma(mt_0/2^k) = e^{(mt_0/2^k)X}$ for all $m \in \mathbb{Z}$, $k \in \mathbb{N}$; dyadic multiples of t_0 are dense and both sides continuous, so $\gamma(t) = e^{tX}$ for all t . Differentiating at 0 gives uniqueness. ■

Remark 1.18 (A first miracle, and a preview of Stone). Mere continuity forced analyticity — continuous symmetry is automatically differentiable symmetry, the matrix-sized shadow of Hilbert’s fifth problem. The infinite-dimensional sibling, *Stone’s theorem* ($U(t) = e^{-itA}$, A self-adjoint), runs quantum time evolution and is quoted in Lecture 9; ours is its finite-dimensional case, the licence to speak of *the generator* of a continuous symmetry.

A nonzero one-parameter generator can act on space in sharply different ways. On the plane, the nonzero elements of $\mathfrak{sl}(2, \mathbb{R})$ fall into three qualitative generator/flow types — rotations, squeezes, and shears — fixed by the eigenvalues of X (Figure 1.2); the zero generator gives the stationary identity flow. This is not a claim that there are only three conjugacy classes of elements or subgroups; exceptional parameter values and sign/orientation data must also be retained. The squeezes return as Lorentz boosts in Lecture 14.

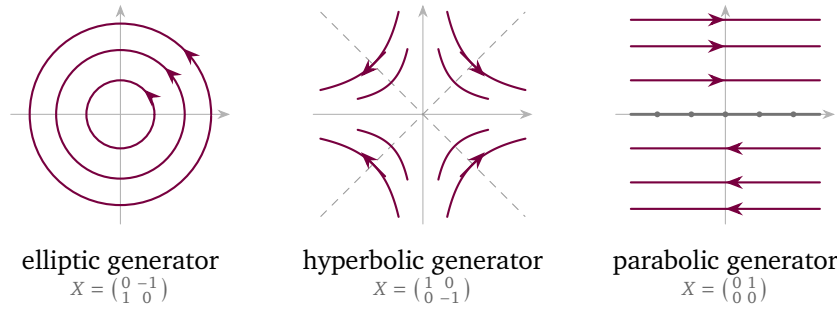


Figure 1.2: Three qualitative generator types in $\mathfrak{sl}(2, \mathbb{R})$, drawn through their flows $t \mapsto e^{tX}$ on the plane. For the displayed elliptic representative, $\text{tr}(e^{tX}) = 2 \cos t$: it reaches the central elements $\pm \mathbb{I}$ when $t \in \pi\mathbb{Z}$ and has $|\text{tr}(e^{tX})| < 2$ otherwise. For the displayed hyperbolic representative, $\text{tr}(e^{tX}) = 2 \cosh t$: it is $\mathbb{I}$ at $t = 0$ and has trace greater than 2 otherwise. For the displayed parabolic representative, $e^{tX} = \mathbb{I} + tX$ has trace 2; it is the identity at $t = 0$ and a nontrivial shear otherwise. Thus these pictures organise flow behaviour, not the individual conjugacy classes: the zero generator and central group elements are exceptional, each qualitative type contains further conjugacy data, and signs/orientations — notably X versus $-X$ for a nilpotent generator inside $\text{SL}(2, \mathbb{R})$ — are not encoded by the drawings.

Theorem 1.19 (Lie product formula). For all $X, Y \in M_n(\mathbb{C})$,
$$e^{X+Y} = \lim_{m \rightarrow \infty} \left(e^{X/m} e^{Y/m} \right)^m.$$

Proof. Multiplying series, $e^{X/m} e^{Y/m} = \mathbb{I} + \frac{X+Y}{m} + R_m$ with $\|R_m\| \leq c/m^2$, $c = c(X, Y)$: every term of combined degree $d \geq 2$ carries m^{-d} , with coefficients dominated by those of $e^{\|X\|} e^{\|Y\|}$. For large m this lies in the domain of $\log$, and the tail estimate $\|\log(\mathbb{I} + B) - B\| \leq \|B\|^2$ for $\|B\| \leq \frac{1}{2}$ gives $\log(e^{X/m} e^{Y/m}) = \frac{X+Y}{m} + S_m$ with $\|S_m\| \leq c'/m^2$. Hence, by $(e^Z)^m = e^{mZ}$ and continuity of $\exp$,

$$\left(e^{X/m} e^{Y/m} \right)^m = \exp(X + Y + mS_m) \xrightarrow{m \rightarrow \infty} e^{X+Y}.$$

■

(The formula is modern technology: alternating many short steps of e^{-itH_1} and e^{-itH_2} is how quantum simulators enact $e^{-it(H_1+H_2)}$, at the $O(1/m)$ rate visible in the proof.)

1.4 The Lie algebra of a matrix Lie group

Here is the central definition of Part II. It asks of a matrix not where it is, but where it can take you.

Definition 1.20 (Lie algebra of a matrix Lie group). The Lie algebra of a matrix Lie group $G \subseteq \mathrm{GL}(n, \mathbb{C})$ is

$$\mathfrak{g} = \mathrm{Lie}(G) = \{X \in M_n(\mathbb{C}) : e^{tX} \in G \text{ for all } t \in \mathbb{R}\},$$

the set of matrices whose entire one-parameter flow stays inside G . Lie algebras wear fraktur: $\mathrm{Lie}(\mathrm{SU}(n)) = \mathfrak{su}(n)$, and so on.

Note the *real* parameter t , whatever the field of the matrices — a deliberate choice sorted out in Section 1.6. Physicists call elements of $\mathfrak{g}$ *infinitesimal generators*; Corollary 1.32 will justify the name. Since every exponential is invertible, $\mathfrak{gl}(n, \mathbb{C}) = M_n(\mathbb{C})$ and $\mathfrak{gl}(n, \mathbb{R}) = M_n(\mathbb{R})$: no condition at all. For the rest, one two-step pattern does everything: *differentiate the defining constraint at $t = 0$ for necessity, exponentiate the linearised constraint for sufficiency*.

Example 1.21 ($\mathfrak{sl}$: trace zero). $X \in \mathfrak{sl}(n, \mathbb{C})$ means $\det e^{tX} = e^{t \mathrm{tr} X} = 1$ for all t (1.5), i.e. $\mathrm{tr} X = 0$ (differentiate at $t = 0$); the converse is immediate. So $\mathfrak{sl}(n, \mathbb{C}) = \{X : \mathrm{tr} X = 0\}$, likewise $\mathfrak{sl}(n, \mathbb{R})$ over $\mathbb{R}$.

Example 1.22 ($\mathfrak{u}$ and $\mathfrak{su}$: anti-Hermitian). $X \in \mathfrak{u}(n)$ means $(e^{tX})^\dagger e^{tX} = e^{tX^\dagger} e^{tX} = \mathbb{I}$ for all t ; the derivative at $t = 0$ (product rule, Proposition 1.12(v)) forces $X^\dagger + X = 0$, and conversely $X^\dagger = -X$ gives $(e^{tX})^\dagger e^{tX} = e^{-tX^\dagger} e^{tX} = \mathbb{I}$. So $\mathfrak{u}(n)$ is the space of *anti-Hermitian* matrices and $\mathfrak{su}(n) = \{X^\dagger = -X, \mathrm{tr} X = 0\}$. Counting real parameters (imaginary diagonal, free complex entries above it) gives $\dim_{\mathbb{R}} \mathfrak{u}(n) = n^2$ and, the trace here being one real condition, $\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$. For $n = 1$: $\mathfrak{u}(1) = i\mathbb{R}$, the tangent line of Figure 1.1.

Example 1.23 ($\mathfrak{so}$: antisymmetric — and blind). The same computation with T for $\dagger$ gives $\mathrm{Lie}(\mathrm{O}(n)) = \{X \in M_n(\mathbb{R}) : X^T = -X\}$, of dimension $n(n-1)/2$. An antisymmetric matrix has zero diagonal, so $\mathrm{tr} X = 0$ is automatic and the determinant condition adds nothing: $\mathrm{Lie}(\mathrm{O}(n)) = \mathrm{Lie}(\mathrm{SO}(n)) = \mathfrak{so}(n)$. The algebra cannot tell $\mathrm{O}(n)$ from $\mathrm{SO}(n)$ — a moral, not an accident: differentiation at the identity sees only the connected piece containing it — $\mathrm{SO}(n)$, never the reflections.

Example 1.24 (The Lorentz algebra). For $\mathrm{O}(1, 3)$ the constraint $(e^{tX})^T \eta e^{tX} = \eta$ linearises to $X^T \eta + \eta X = 0$, i.e. ηX antisymmetric: in 1+3 block form, $X = \begin{pmatrix} 0 & \vec{b}^T \\ \vec{b} & A \end{pmatrix}$ with $\vec{b} \in \mathbb{R}^3$ free and $A^T = -A$, so $\dim_{\mathbb{R}} \mathfrak{so}(1, 3) = 3 + 3 = 6$: three boosts ($\vec{b}$, symmetric placement), three rotations (A , antisymmetric). How profoundly the two triples differ is Lecture 14's business.

Example 1.25 (The Heisenberg algebra). For H_3 , exponentials of strictly upper triangular X truncate to unitriangular polynomials, and differentiating the fixed entries of e^{tX} at $t = 0$ forces the converse: $\mathfrak{h}_3$ is the three-dimensional space of strictly upper triangular matrices,

with $[X, X'] = (ab' - a'b)E_{13}$ — every commutator points along the corner entry, which commutes with everything: the shadow of $[\hat{q}, \hat{p}] = i\hbar$ (Lecture 13).

So far $\mathfrak{g}$ is only a set. The next theorem gives it its structure — and reveals why the definition of a matrix Lie group insisted on closedness.

Theorem 1.26 ($\mathfrak{g}$ is closed under sums, conjugation, commutators). *Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$. Then*

- (i) $\mathfrak{g}$ is a real vector subspace of $M_n(\mathbb{C})$;
- (ii) $gXg^{-1} \in \mathfrak{g}$ for all $g \in G, X \in \mathfrak{g}$;
- (iii) $[X, Y] := XY - YX \in \mathfrak{g}$ for all $X, Y \in \mathfrak{g}$.

Proof. (i) $e^{t(sX)} = e^{(ts)X} \in G$ handles real multiples. For sums, the Lie product formula gives $e^{t(X+Y)} = \lim_m (e^{tX/m} e^{tY/m})^m$: every matrix under the limit lies in G , the limit is invertible (an exponential), and G is closed in $GL(n, \mathbb{C})$ — so $e^{t(X+Y)} \in G$ for all t , i.e. $X+Y \in \mathfrak{g}$. Here the closedness hypothesis earns its keep; it will once more, in Theorem 1.31.

(ii) $e^{t(gXg^{-1})} = g e^{tX} g^{-1} \in G$ (Proposition 1.12(ii)).

(iii) By (ii), the curve $Z(t) = e^{tX} Y e^{-tX}$ lies in $\mathfrak{g}$ for all t ; by (i), $\mathfrak{g}$ is a finite-dimensional real subspace of $M_n(\mathbb{C})$, hence a closed subset. The difference quotients $(Z(t) - Z(0))/t$ lie in $\mathfrak{g}$ and converge (product rule) to $Z'(0) = XY - YX = [X, Y]$, which therefore lies in $\mathfrak{g}$. ■

The bracket $[X, Y]$ measures noncommutativity to first order; that it never leads out of $\mathfrak{g}$ is the linearised residue of G being closed under multiplication. The structure deserves axioms.

Definition 1.27 (Lie algebra; homomorphism). *A Lie algebra over $\mathbb{R}$ (or $\mathbb{C}$) is a real (complex) vector space $\mathfrak{g}$ with a bracket $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$, bilinear, antisymmetric, and satisfying the Jacobi identity*

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

A linear map with $\varphi([X, Y]) = [\varphi(X), \varphi(Y)]$ is a *homomorphism* of Lie algebras, a bijective one an *isomorphism* ($\cong$). Relative to a basis $X_1, \dots, X_d$, the *structure constants* f_{ab}^c defined by $[X_a, X_b] = f_{ab}^c X_c$ (sum over c) pin the algebra down completely.

The matrix commutator is bilinear and antisymmetric, and the twelve terms of its Jacobi identity cancel in pairs upon expansion: every $\text{Lie}(G)$ is a real Lie algebra in the abstract sense.

Example 1.28 ($\mathfrak{su}(2)$ and $\mathfrak{so}(3)$). Two three-dimensional specimens will dominate the next four lectures. The anti-Hermitian traceless matrices $X_a = -\frac{i}{2}\sigma_a$ form a basis of $\mathfrak{su}(2)$, and (1.3) gives $[\sigma_j, \sigma_k] = 2i\epsilon_{jkl}\sigma_l$, hence structure constants $f_{jk}^l = \epsilon_{jkl}$:

$$[X_j, X_k] = -\frac{1}{4}[\sigma_j, \sigma_k] = \epsilon_{jkl} X_l. \quad (1.6)$$

In $\mathfrak{so}(3)$, for $\vec{\omega} \in \mathbb{R}^3$ let $\hat{\omega}$ have entries $(\hat{\omega})_{jk} = -\epsilon_{ajk}\omega_a$, so that $\hat{\omega} \vec{x} = \vec{\omega} \times \vec{x}$: the *hat map* $\vec{\omega} \mapsto \hat{\omega}$ is a linear bijection $\mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ (both have dimension 3), and the basis $L_a = \hat{e}_a$ generates the rotations about the axes — $e^{\theta L_3}$ is Example 1.13's $R(\theta)$ in the xy -plane.

Proposition 1.29 (One algebra, three costumes). *The hat map and the assignment $X_a \mapsto L_a$ are Lie algebra isomorphisms*

$$(\mathbb{R}^3, \times) \cong \mathfrak{so}(3) \cong \mathfrak{su}(2).$$

Proof. The cross product is bilinear, antisymmetric, and its Jacobi identity is the classical identity $\vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a}) + \vec{c} \times (\vec{a} \times \vec{b}) = 0$. The same identity, applied to a vector $\vec{x}$, evaluates the matrix commutator:

$$[\hat{\omega}, \hat{\nu}] \vec{x} = \vec{\omega} \times (\vec{\nu} \times \vec{x}) - \vec{\nu} \times (\vec{\omega} \times \vec{x}) = (\vec{\omega} \times \vec{\nu}) \times \vec{x},$$

so $[\hat{\omega}, \hat{\nu}] = \widehat{\vec{\omega} \times \vec{\nu}}$ and the hat map preserves brackets. In particular $[L_j, L_k] = \widehat{e_j \times e_k} = \epsilon_{jkl} L_l$ — the structure constants (1.6) — so $X_a \mapsto L_a$ extends to an isomorphism too. ■

Remark 1.30 (The algebra is short-sighted). So $\mathfrak{su}(2) \cong \mathfrak{so}(3)$, yet the groups are not isomorphic: $SU(2)$ is the sphere S^3 of Example 1.7, while $SO(3)$, Lecture 11 will show, is that sphere with antipodes glued — with consequences the size of the electron's spin. Together with $\text{Lie}(O(n)) = \text{Lie}(SO(n))$, this calibrates how much the algebra remembers: everything near the identity, nothing global.

Table 1.1 collects the lecture's bookkeeping.

G	condition	$\mathfrak{g} = \text{Lie}(G)$	$\dim_{\mathbb{R}} \mathfrak{g}$	cpt.	conn.
$GL(n, \mathbb{C})$	$\det A \neq 0$	$M_n(\mathbb{C})$	$2n^2$	no	yes
$SL(n, \mathbb{C})$	$\det A = 1$	$\text{tr } X = 0$	$2n^2 - 2$	no	yes
$GL(n, \mathbb{R})$	real, $\det A \neq 0$	$M_n(\mathbb{R})$	n^2	no	no (2)
$SL(n, \mathbb{R})$	real, $\det A = 1$	$\text{tr } X = 0$	$n^2 - 1$	no	yes
$O(n)$	$A^T A = \mathbb{I}$	$X^T = -X$	$\frac{n(n-1)}{2}$	yes	no (2)
$SO(n)$	$\dots$, $\det A = 1$	$X^T = -X$	$\frac{n(n-1)}{2}$	yes	yes
$U(n)$	$A^\dagger A = \mathbb{I}$	$X^\dagger = -X$	n^2	yes	yes
$SU(n)$	$\dots$, $\det A = 1$	$\dots$, $\text{tr } X = 0$	$n^2 - 1$	yes	yes
$O(1, 3)$	$A^T \eta A = \eta$	$X^T \eta = -\eta X$	6	no	no (4)
H_3	unitriangular	strictly upper	3	no	yes

Table 1.1: The zoo at a glance ($n \geq 2$ where it matters). Disconnectedness: Proposition 1.9; connectedness: Section 1.5 (unitary, orthogonal), Sheet 8 (linear), Lecture 13 (H_3), Lecture 14 (the four Lorentz components).

1.5 The exponential as a bridge

By construction $\exp$ maps $\mathfrak{g}$ into G , and by Theorem 1.16 it is injective near 0. What is not obvious — and is the deepest statement of this lecture — is that near the identity it *fills* G : a closed subgroup of $GL(n, \mathbb{C})$ has no elements loitering near $\mathbb{I}$ other than exponentials of its own Lie algebra.

Theorem 1.31 (Exponential coordinates). *Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$. There exist $\varepsilon \in (0, \log 2)$ and a neighbourhood V of $\mathbb{I}$ in $GL(n, \mathbb{C})$ such that every $g \in G \cap V$ is $g = e^X$ for a unique $X \in \mathfrak{g}$ with $\|X\| < \varepsilon$.*

Proof. Choose a real subspace D with $M_n(\mathbb{C}) = \mathfrak{g} \oplus D$ and define $\varphi: \mathfrak{g} \times D \rightarrow GL(n, \mathbb{C})$, $\varphi(X, Y) = e^X e^Y$ — smooth (convergent power series), with $\varphi(X, Y) = \mathbb{I} + X + Y + (\text{higher order})$, so its derivative at $(0, 0)$ is $(X, Y) \mapsto X + Y$, a linear bijection onto $M_n(\mathbb{C})$. By the inverse function theorem (Remark 1.3) there are $\delta > 0$ and a neighbourhood V_0 of $\mathbb{I}$ in which every g factors *uniquely* as $g = e^X e^Y$, $X \in \mathfrak{g}$, $Y \in D$, $\|X\|, \|Y\| < \delta$, the factors depending continuously on g . It remains to show that for $g \in G$ near $\mathbb{I}$ the second factor vanishes. Suppose not:

there are $g_m \in G$, $g_m \rightarrow \mathbb{K}$, with $g_m = e^{X_m} e^{Y_m}$, $Y_m \neq 0$, and $X_m, Y_m \rightarrow 0$ by continuity; since $e^{X_m} \in G$, also $e^{Y_m} = e^{-X_m} g_m \in G$. The unit vectors $Y_m / \|Y_m\|$ live on the unit sphere of D , which is compact; after passing to a subsequence, $Y_m / \|Y_m\| \rightarrow Y \in D$, $\|Y\| = 1$. Fix $t \in \mathbb{R}$ and let k_m be the integer nearest $t / \|Y_m\|$, so $k_m \|Y_m\| \rightarrow t$ (rounding error $\leq \|Y_m\| \rightarrow 0$); then

$$(e^{Y_m})^{k_m} = \exp\left(k_m \|Y_m\| \frac{Y_m}{\|Y_m\|}\right) \xrightarrow{m \rightarrow \infty} e^{tY}.$$

Every term on the left lies in G , the limit is invertible, and G is closed in $\mathrm{GL}(n, \mathbb{C})$ — its second decisive appearance — so $e^{tY} \in G$ for every t . That says $Y \in \mathfrak{g}$; but $Y \in D$ and $\mathfrak{g} \cap D = \{0\}$, contradicting $\|Y\| = 1$. Hence some $V \subseteq V_0$ has $G \cap V \subseteq \exp(\mathfrak{g})$; shrinking $\varepsilon \leq \delta$ below $\log 2$ makes X unique (Theorem 1.16). ■

Corollary 1.32 (Generators are velocities). $\mathfrak{g}$ consists exactly of the velocities $\gamma'(0)$ of the curves $\gamma: (-\delta, \delta) \rightarrow G$ that are differentiable at 0 with $\gamma(0) = \mathbb{K}$.

Proof. ($\subseteq$) X is the velocity of e^{tX} . ($\supseteq$) For small t , Theorem 1.31 writes $\gamma(t) = e^{Z(t)}$, $Z(t) = \log \gamma(t) \in \mathfrak{g}$; with $A = \gamma'(0)$, the tail estimate $\|\log(\mathbb{K} + B) - B\| \leq \|B\|^2$ gives $Z(t) = tA + o(t)$, so $Z(t)/t \rightarrow A$ inside the closed subspace $\mathfrak{g}$. ■

The corollary is why “infinitesimal generator” is honest language — and mechanics has used it all along: for a rigid-body attitude $t \mapsto R(t) \in \mathrm{SO}(3)$, the curve $t \mapsto R(s+t)R(s)^{-1}$ passes through $\mathbb{K}$, so $\dot{R}(s)R(s)^{-1} = \hat{\omega}(s) \in \mathfrak{so}(3)$ and body points obey $\dot{\vec{x}} = \vec{\omega} \times \vec{x}$: the angular velocity vector is a Lie algebra element, named in mechanics before its time.

How much of G does $\exp$ reach beyond a neighbourhood of $\mathbb{K}$? Exactly the part connected to it. Write G^0 for the *identity component* of G : the set of $g \in G$ joined to $\mathbb{K}$ by some continuous path in G .

Theorem 1.33 (The identity component). G^0 is a normal subgroup of G , and $G^0 = \{e^{X_1} \cdots e^{X_k} : k \in \mathbb{N}, X_i \in \mathfrak{g}\}$.

Proof. If γ_g, γ_h join $\mathbb{K}$ to g, h , then $\gamma_g \gamma_h$ joins it to gh and γ_g^{-1} to g^{-1} : G^0 is a subgroup; conjugation by any $a \in G$ is continuous and fixes $\mathbb{K}$, hence permutes such paths: $aG^0 a^{-1} = G^0$.

Let P be the set of finite products of exponentials — a subgroup, since $(e^X)^{-1} = e^{-X}$. Shrinking the factors, $t \mapsto e^{tX_1} \cdots e^{tX_k}$ is a path from $\mathbb{K}$, so $P \subseteq G^0$. Conversely, $W = G \cap V$ of Theorem 1.31 satisfies $W \subseteq \exp(\mathfrak{g}) \subseteq P$ and is a neighbourhood of $\mathbb{K}$ in G ; for $p \in P$, pW is a neighbourhood of p contained in P , so P is open in G ; each coset aP is open, so the complement of P (a union of cosets) is open too. P is thus open and closed in G ; since G^0 is connected and meets P in $\mathbb{K}$, $G^0 \subseteq P$. ■

Corollary 1.34. A connected matrix Lie group is path-connected and is generated by any neighbourhood of its identity. G^0 is itself a matrix Lie group with $\mathrm{Lie}(G^0) = \mathfrak{g}$, and G/G^0 is the group of components: e.g. $\det: \mathrm{O}(n) \rightarrow \{\pm 1\}$ has kernel $\mathrm{SO}(n) = \mathrm{O}(n)^0$, so $\mathrm{O}(n)/\mathrm{SO}(n) \cong \mathbb{Z}_2$ by Lecture 2’s first isomorphism theorem.

Proof. If G is connected the proof above gives $G = P$, with W shrunk into any given neighbourhood — the generation claim — and every element of P is path-connected to $\mathbb{K}$. G^0 is closed in G (its complement is a union of cosets of an open subgroup), hence a matrix Lie group containing the flows e^{tX} : $\mathrm{Lie}(G^0) = \mathfrak{g}$. The example is the first isomorphism theorem applied to $\det$. ■

For the compact members of the zoo, the linear-algebra prerequisite upgrades “generated by exponentials” to “equal to the exponentials”.

Theorem 1.35 (Surjectivity for the unitary and rotation groups). $\exp$ maps $\mathfrak{u}(n)$ onto $U(n)$, $\mathfrak{su}(n)$ onto $SU(n)$, and $\mathfrak{so}(n)$ onto $SO(n)$; consequently these three groups are (path-)connected.

Proof. $U(n)$: the spectral theorem (Remark 1.3) writes $U = VDV^\dagger$ with V unitary and $D = \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_n})$; then $X = V \text{diag}(i\theta_j)V^\dagger$ is anti-Hermitian with $e^X = U$.

$SU(n)$: in addition $\det U = e^{i\sum \theta_j} = 1$, so $\sum_j \theta_j = 2\pi k$; replacing θ_1 by $\theta_1 - 2\pi k$ changes neither $e^{i\theta_1}$ nor U and puts X in $\mathfrak{su}(n)$.

$SO(n)$: the normal form for special orthogonal matrices (Remark 1.3) writes $R = QBQ^\top$ with $Q \in O(n)$ and $B = \text{blockdiag}(R(\theta_1), \dots, R(\theta_m), \mathbb{K})$: eigenvalue pairs $e^{\pm i\theta}$ give rotation blocks, and the eigenvalue -1 , occurring an even number of times since $\det R = 1$, pairs into blocks $R(\pi)$. Then $X = Q \text{blockdiag}(\theta_1 J, \dots, \theta_m J, 0)Q^\top$ is antisymmetric with $e^X = R$, by Example 1.13 applied blockwise.

Connectedness: $t \mapsto e^{tX}$ is a path from $\mathbb{K}$ to e^X inside the group. ■

This settles three “yes” entries of Table 1.1: for these groups one exponential suffices in Theorem 1.33 — a luxury of compactness, not a law:

Example 1.36 (exp need not be surjective). $A = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \in \text{SL}(2, \mathbb{C})$ is no exponential of $\mathfrak{sl}(2, \mathbb{C})$: such an X has eigenvalues $\pm\lambda$ (their sum is $\text{tr} X = 0$), so e^X has eigenvalues $e^{\pm\lambda}$, while both eigenvalues of A are -1 : then $\lambda \in i\pi(2\mathbb{Z}+1)$, so $\lambda \neq -\lambda$, X is diagonalisable, and so is e^X — but A , with one-dimensional (-1) -eigenspace, is not. Yet $\text{SL}(2, \mathbb{C})$ is connected (Sheet 8), and two exponentials reach A : $A = \exp(i\pi \text{diag}(1, -1)) \exp\begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$, both exponents traceless, in keeping with Theorem 1.33.

Let us take stock. Locally the bridge is perfect: Theorems 1.16 and 1.31 say that near the identity G is $\mathfrak{g}$ in curved coordinates; the Baker–Campbell–Hausdorff formula (stated in Lecture 9) even expresses $\log(e^X e^Y)$ in iterated brackets alone. Globally, three phenomena appeared: $\exp$ can fail to be injective (the circle), can fail to be surjective ($\text{SL}(2, \mathbb{C})$), and distinct groups can share one algebra ($\mathfrak{su}(2) \cong \mathfrak{so}(3)$). The first two are bookkeeping; the third is a door, and Lectures 9 and 11 walk through it. (Sheet 8 adds the polar decomposition of $\text{GL}(n, \mathbb{C})$, anatomy Lecture 14 will need.)

1.6 Real and complex Lie algebras, and the physicist’s conventions

One structural point was left deliberately loose, and physics conventions live in the slack. The Lie algebras of Definition 1.20 are *real* vector spaces — the flow parameter is real — even when their matrices are complex. Some are complex spaces too ($\mathfrak{gl}(n, \mathbb{C})$, $\mathfrak{sl}(n, \mathbb{C})$) are closed under multiplication by i ; the most important are not, since $X^\dagger = -X$ gives $(iX)^\dagger = +iX$: multiplying an anti-Hermitian matrix by i produces a *Hermitian* one. “ $\mathfrak{su}(2)$ consists of complex matrices” is not “ $\mathfrak{su}(2)$ is a complex Lie algebra”. When complex scalars are wanted — Lecture 10 will want them badly — they must be adjoined by hand.

Definition 1.37 (Complexification). Let $\mathfrak{g} \subseteq M_n(\mathbb{C})$ be a real Lie algebra of matrices with $\mathfrak{g} \cap i\mathfrak{g} = \{0\}$. Its *complexification* is the complex Lie algebra

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \oplus i\mathfrak{g} = \{X + iY : X, Y \in \mathfrak{g}\},$$

with the bracket extended by complex bilinearity. (For $\mathfrak{g}$ already complex the recipe degenerates; the abstract $\mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$ then *doubles* the algebra — pedantry turned physics in Lecture 14, where complexifying $\mathfrak{so}(1, 3)$ yields *two* commuting copies of $\mathfrak{sl}(2, \mathbb{C})$: the left/right labels on Weyl spinors.)

Proposition 1.38 (The classical complexifications). $\mathfrak{su}(n)_{\mathbb{C}} = \mathfrak{sl}(n, \mathbb{C})$, $\mathfrak{u}(n)_{\mathbb{C}} = \mathfrak{gl}(n, \mathbb{C})$, $\mathfrak{sl}(n, \mathbb{R})_{\mathbb{C}} = \mathfrak{sl}(n, \mathbb{C})$.

Proof. Anti-Hermitian $\cap$ Hermitian = $\{0\}$ gives $\mathfrak{su}(n) \cap i\mathfrak{su}(n) = \{0\}$, and any traceless M splits as

$$M = \underbrace{\frac{1}{2}(M - M^\dagger)}_{=:X} + i \underbrace{\left(-\frac{i}{2}\right)(M + M^\dagger)}_{=:Y},$$

with X, Y anti-Hermitian and traceless (as $\text{tr } M^\dagger = \overline{\text{tr } M} = 0$): the first claim; dropping traces gives the second, real/imaginary parts the third. ■

A complexification forgets information: different real algebras can share one. The pair that matters most in physics:

Proposition 1.39 (Two real forms of $\mathfrak{sl}(2, \mathbb{C})$). $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{R})_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{C})$, but $\mathfrak{su}(2) \not\cong \mathfrak{sl}(2, \mathbb{R})$ as real Lie algebras.

Proof. The complexifications are Proposition 1.38; for the difference, equip $\mathfrak{su}(2)$ with the real inner product $\langle A, B \rangle = -\text{tr}(AB)$; by cyclicity of the trace $\langle [X, A], B \rangle + \langle A, [X, B] \rangle = -\text{tr}(XAB - ABX) = 0$, so every $\text{ad}_X = [X, \cdot]$ is skew-symmetric and has purely imaginary eigenvalues (linear algebra). In $\mathfrak{sl}(2, \mathbb{R})$, by contrast, $H = \text{diag}(1, -1)$ and the off-diagonal units E, F obey $[H, E] = 2E$, $[H, F] = -2F$, so ad_H has the real eigenvalues 2, -2, 0. An isomorphism φ would intertwine ad_H with $\text{ad}_{\varphi(H)}$, forcing real nonzero eigenvalues on an ad of $\mathfrak{su}(2)$ — impossible. ■

Every ad skew versus some ad stretching: precisely Lecture 14's contrast between rotations and boosts — one complex algebra, several real forms, different physics.

1.6.1 The physicist's conventions

Physics texts populate $\mathfrak{su}(n)$ not with anti-Hermitian matrices but with *Hermitian generators* — Hermitian operators are observables. The translation is one factor of i : for a basis $X_1, \dots, X_d$ of $\mathfrak{g} \subseteq \mathfrak{u}(n)$ with real structure constants f_{ab}^c , set $T_a = iX_a$ (Hermitian); then

$$e^{\theta^a X_a} = e^{-i\theta^a T_a}, \quad [T_a, T_b] = -[X_a, X_b] = if_{ab}^c T_c : \quad (1.7)$$

the mathematician's (X_a, f_{ab}^c) and the physicist's (T_a, if_{ab}^c) are one structure in two fonts. The sign $e^{-i\theta T}$ matches the quantum-mechanical $e^{-itH/\hbar}$ and is fixed henceforth.

Example 1.40 (Angular momentum, before quantum mechanics). For $\mathfrak{su}(2)$ with $X_a = -\frac{i}{2}\sigma_a$, the Hermitian generators are $J_a = \frac{1}{2}\sigma_a$, and (1.6), (1.4) become

$$[J_j, J_k] = i\epsilon_{jkl} J_l, \quad e^{-i\theta \vec{n} \cdot \vec{J}} = \cos \frac{\theta}{2} \mathbb{I} - i \sin \frac{\theta}{2} \vec{n} \cdot \vec{\sigma}. \quad (1.8)$$

The commutation relations of angular momentum, derived from the geometry of $\text{SU}(2)$ alone: Lecture 10 takes (1.8) as the seed from which the entire theory of spin grows.

Remark 1.41 (Where the Hermitian generators actually live). The real span ig of the T_a is *not* bracket-closed ($[iX, iY] = -[X, Y] \in \mathfrak{g}$ — the stray i in (1.7)): “the algebra of Hermitian generators” is shorthand for $\mathfrak{g}$ or $\mathfrak{g}_{\mathbb{C}}$, where X_a, T_a and Lecture 10's ladder operators $\sigma_{\pm} = \frac{1}{2}(\sigma_1 \pm i\sigma_2)$ all coexist. Complexify, and the bookkeeping dissolves.

The bridge is complete: groups, algebras, the exponential between. Lecture 9 sends Part I's representations across it — differentiating, exponentiating back when topology permits, meeting Noether's theorem where the directions touch physics; the rest of Part I survives once averaging becomes integration (Lecture 12).

Remark 1.42 (Computer algebra). The notebook computes exponentials by truncated series and by `MatrixExp`; checks $e^X \in \mathrm{SU}(2)$ for random $X \in \mathfrak{su}(2)$; plots one-parameter orbits in $\mathrm{SL}(2, \mathbb{R})$ (ellipses, hyperbolas, shears — the flows of Figure 1.2, met again as boosts in Lecture 14); measures the $O(1/m)$ Lie-product rate; and recovers $\mathfrak{so}(3)$'s structure constants numerically.

Summary.

- *Matrix Lie group.* A closed subgroup of $\mathrm{GL}(n, \mathbb{C})$ (Definition 1.1); the classical groups qualify, with compactness and connectedness as catalogued (Proposition 1.4).
- *Matrix exponential.* $e^X = \sum_k X^k/k!$ with $\det e^X = e^{\mathrm{tr} X}$ (Theorem 1.15), a local inverse to log near the identity (Theorem 1.16).
- *One-parameter subgroups.* Every continuous $\mathbb{R} \rightarrow G$ has the form $t \mapsto e^{tX}$ (Theorem 1.17); the Lie product formula rebuilds products from generators (Theorem 1.19).
- *Lie algebra.* $\mathfrak{g} = \{X : e^{tX} \in G \ \forall t\}$, a real vector space closed under the commutator (Definition 1.20, Theorem 1.26); its elements are tangent velocities (Corollary 1.32).
- *Exponential and complexification.* $\exp$ is onto for $\mathrm{U}(n)$, $\mathrm{SU}(n)$, $\mathrm{SO}(n)$ (Theorem 1.35); $\mathfrak{sl}(2, \mathbb{C})$ has real forms $\mathfrak{su}(2)$ and $\mathfrak{sl}(2, \mathbb{R})$ (Proposition 1.39); physics dictionary $X = -iA$, A Hermitian (eq. (1.7)).

Tutorial. Exercise Sheet 8.

References

Hall Ch. 1–2, §§3.1–3.4, 3.6; Stillwell Ch. 1–2 (quaternionic $\mathrm{SU}(2)$); Erdmann–Wildon Ch. 1 (abstract Lie algebras); Jeevanjee (physics notation); Gilmore Ch. 1–3 (background reading).

Bibliography

The per-lecture *References* sections cite the works below by author and short title; full details are collected here. Chapter and section numbers in the lecture references follow these editions.

- M. A. Armstrong, *Groups and Symmetry*. Springer (UTM), 1988.
- V. Bargmann, On unitary ray representations of continuous groups. *Annals of Mathematics* **59** (1954), 1–46.
- A. O. Barut and R. Rączka, *Theory of Group Representations and Applications*, 2nd ed. World Scientific, 1986.
- R. Berndt, *Representations of Linear Groups: An Introduction Based on Examples from Physics and Number Theory*. Vieweg, 2007. DOI: 10.1007/978-3-8348-9401-4. Theorem 5.2, p. 71 gives the SNAG theorem used for the translation subgroup.
- D. M. Bishop, *Group Theory and Chemistry*. Dover, 1993 (orig. Oxford University Press, 1973).
- O. Bratteli and D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics 2: Equilibrium States, Models in Quantum Statistical Mechanics*, 2nd ed. (1997), 2nd printing. Springer, 2002. Section 5.2.1, pp. 7–8 gives Fock-space second quantisation, its finite-particle analytic core and closure, and the lifted unitary group.
- G. Costa and G. Fogli, *Symmetries and Group Theory in Particle Physics*. Springer (Lecture Notes in Physics 823), 2012.
- J. B. Conway, *A Course in Functional Analysis*, 1st ed. Springer (GTM 96), 1985. Chapter IX, Theorems 6.4 and 6.6 give the strong/weak bicommutant closure and the multiplication maximal-abelian theorem for sigma-finite measure spaces.
- J. P. Elliott and P. G. Dawber, *Symmetry in Physics*, 2 vols. Macmillan, 1979.
- K. Erdmann and M. J. Wildon, *Introduction to Lie Algebras*. Springer (SUMS), 2006.
- W. Fulton and J. Harris, *Representation Theory: A First Course*. Springer (GTM 129), 1991.
- G. B. Folland, *Harmonic Analysis in Phase Space*. Princeton University Press (Annals of Mathematics Studies 122), 1989. Chapter 1, Sections 2–5, including Theorem 1.50, treats the Heisenberg group, Schrödinger/Weyl representation, and Stone–von Neumann theorem.
- G. B. Folland, *A Course in Abstract Harmonic Analysis*, 2nd ed. Chapman & Hall/CRC, 2016. Section 6.6, Theorem 6.42, pp. 199–201 gives the regular semidirect-product Mackey theorem, including induced irreducibility, exhaustivity, and equivalence.
- R. Gilmore, *Lie Groups, Lie Algebras, and Some of Their Applications*. Wiley, 1974 (repr. Dover, 2005).
- B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed. Springer (GTM 222), 2015.
- N. Jeevanjee, *An Introduction to Tensors and Group Theory for Physicists*, 2nd ed. Birkhäuser, 2015.
- H. F. Jones, *Groups, Representations and Physics*, 2nd ed. Institute of Physics Publishing,

1998. Chapter 8, Sections 8.3–8.4, pp. 149–164 is the course’s explicit $SU(3)$ further-reading route; Chapter 10, Section 10.4, pp. 208–216 treats the Poincaré little groups; Appendix E, pp. 279–280 derives the covariant on-shell measure from $\theta(p^0)\delta(p^2 - m^2)d^4p$.
- Z.-Q. Ma, *Group Theory for Physicists*. World Scientific, 2007.
 - A. M. L. Messiah and O. W. Greenberg, Symmetrization Postulate and Its Experimental Foundation. *Physical Review* **136** (1964), B248–B267. DOI: <https://doi.org/10.1103/PhysRev.136.B248>.
 - H. Osborn, *Group Theory*. University of Cambridge, Mathematical Tripos Part III lecture notes (available from the DAMTP lecture-notes pages).
 - M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, rev. ed. Academic Press, 1980. Chapter VIII, Theorems VIII.5, VIII.8, VIII.10, and VIII.13 and Section VIII.5, Example 2 cover spectral calculus, Stone’s theorem, invariant-domain generators, strong commutation, and the bounded-CCR obstruction.
 - P. Ramond, *Group Theory: A Physicist’s Survey*. Cambridge University Press, 2010. Chapter 11, Section 11.1.1, pp. 224–231 treats the Pauli–Lubanski Casimirs and the massless little group; pp. 230–231 distinguish finite helicity from continuous spin and record the two integer/half-odd-integer cover sectors.
 - H. Rauch, A. Zeilinger, G. Badurek, A. Wilfing, W. Bauspiess, and U. Bonse, Verification of Coherent Spinor Rotation of Fermions. *Physics Letters A* **54** (1975), 425–427. DOI: [10.1016/0375-9601\(75\)90798-7](https://doi.org/10.1016/0375-9601(75)90798-7).
 - J. Schwichtenberg, *Physics from Symmetry*, 2nd ed. Springer, 2018.
 - J.-P. Serre, *Linear Representations of Finite Groups*. Springer (GTM 42), 1977.
 - J. Stillwell, *Naive Lie Theory*. Springer (UTM), 2008.
 - E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*. Princeton University Press (Princeton Lectures in Analysis I), 2003. Chapter 3, Section 1, Theorem 1.1, pp. 69–79 gives the mean-square completeness result used in Lecture 12.
 - M. Tinkham, *Group Theory and Quantum Mechanics*. McGraw–Hill, 1964 (repr. Dover, 2003).
 - W.-K. Tung, *Group Theory in Physics*. World Scientific, 1985. Chapter 10, Section 10.4, pp. 191–200 treats induced Poincaré representations, Pauli–Lubanski brackets, and massive and massless little groups.
 - S. Weinberg, *The Quantum Theory of Fields, Vol. I: Foundations*. Cambridge University Press, 1995. Section 5.6 treats causal fields and the relativistic spin–statistics connection, a theorem not derived from a rotation phase in this course.
 - S. A. Werner, R. Colella, A. W. Overhauser, and C. F. Eagen, Observation of the Phase Shift of a Neutron Due to Precession in a Magnetic Field. *Physical Review Letters* **35** (1975), 1053–1055. DOI: [10.1103/PhysRevLett.35.1053](https://doi.org/10.1103/PhysRevLett.35.1053).
 - P. Woit, *Quantum Theory, Groups and Representations: An Introduction*. Springer, 2017.